

Shadman Salif Swanan

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EDUCATION

BRAC University

Bachelor of Science in Computer Science and Engineering (CSE)

2022 – 2026
Dhaka, Bangladesh

- **CGPA:** 3.84 out of 4.00
- **Programming Coursework:** Fundamentals of Programming, Object-Oriented Programming, Data Structures & Algorithms
- **Computational Coursework:** Natural Language Processing, Artificial Intelligence, Computer Network, Linear Algebra, Neural Networks, Digital Logic Design, VLSI Design, Discrete Mathematics

Notre Dame College

Higher Secondary Certificate (HSC), Science

2018 – 2021
Dhaka, Bangladesh

- **GPA:** 5.00 out of 5.00

Ideal School and College

Secondary School Certificate (SSC), Science

2008 – 2018
Dhaka, Bangladesh

- **GPA:** 5.00 out of 5.00 (General Scholarship/ Dhaka Board)

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, PHP

Data Analysis: MySQL, Power BI, Microsoft Excel (Advanced), Process Modeling, EDA

AI/ML Frameworks: PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, Tensorflow, Hugging Face

Web & Tools: React, Vite, Node.js, Express.js, MongoDB, Git & GitHub, Tailwind CSS, Figma

SQA & Testing: Postman (API Testing), Test Case Design

Language Proficiency: English, Bengali

PROJECTS

- **Predictive Telecom Churn Optimization Framework** Engineered a machine learning pipeline processing 100K customer logs with an advanced Gradient Boosting model, achieving a 0.6900 ROC-AUC score and 64.00% churn recall to project an estimated 2.22M USD in annualized retention savings.
- **Real-Time Fraud Detection System** Developed an end-to-end ML pipeline using LightGBM and cost-sensitive threshold tuning to detect credit card anomalies within highly imbalanced transaction streams.
- **Leaf Disease Classification (TinyResNet)** Designed a lightweight convolutional neural network with distortion-aware augmentations, reaching 96.59% classification accuracy under real-world distortions.
- **Token-Level POS Tagging using RNN Models** Implemented sequence labeling frameworks comparing RNN, LSTM, GRU, and BiLSTM architectures with 300D word embeddings to achieve 97% accuracy via BiLSTM.
- **Disease Prediction using Machine Learning** Developed classification models for diabetes prediction using KNN, Decision Tree, and Naive Bayes on clinical data after MinMaxScaler feature mapping.
- **Shikkha.com** Built a full-stack EdTech platform featuring JWT access limits, secure Stripe integration, Automated Testing consoles, and AI-assisted Mistral API educational enhancements.
- **Federal Flight Agency Mission-Critical Aerospace Network Design** Designed a secure and scalable aerospace network architecture using VLSM and floating static routing profiles for automatic mission failover.
- **3D Car Racing Game** Built a 3D racing game using OpenGL with enemy AI, shooting mechanics, weather effects, and camera modes.
- **Automated Sanitizer and Mask Dispensing Robot** Embedded systems project integrating contactless automated sanitizer pumping mechanics and proximity sensors using Arduino Uno.

RESEARCH

Undergraduate Thesis Research | BRAC University

2025 – 2026

Understanding Semantic Drift in Bengali: A Study Based on Diachronic Usage Patterns in Transformer Contexts

- Engineered a comparative embedding framework using GloVe and transformer-based BanglaBERT with an Optuna-optimized hyperparameters across a 120-year corpus to detect true semantic drift with high data processing precision.

Work Experience

Business Analyst Intern

Jun 2026 – Present

Anwar Group of Industries

Dhaka, Bangladesh

- Analyze sales and operational data to identify trends, generate business insights, develop analytical reports, and support data-driven decision-making and process improvement.

GCI World Program

Apr 2026 – July 2026

The University of Tokyo (Matsuo-Iwasawa Laboratory)

Remote

- Participated in an international AI education program focused on machine learning and data science.

CO-CURRICULAR ACTIVITIES

Notre Dame Nature Study Club

Vice President, Department of Photography

- Led photography initiatives and coordinated visual documentation for club activities and events.
- Managed team members and contributed to creative direction and event coverage.

Notre Dame College Model United Nations

Conference Management Officer

- Managed responsibilities in the Conference Management Department during a large-scale academic event.
- Demonstrated strong organizational skills, adaptability, and ability to work efficiently under pressure.

BRAC University Residential Semester

Stage Drama Course Participant

- Completed an intensive stage drama course emphasizing teamwork, communication, and performance skills.

BRAC University Sports Activities

Football Achievements

- Champion & Top Scorer, Intra-Spring 2022 Football Tournament, BRAC University.
- Runners-Up, Crusade 2026 (Semester Tournament) & RS Cup.
- Runners-Up, Residential Semester (RS) Football Tournament.

REFERENCES

Dr. Chowdhury Mofizur Rahman

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